

Cell Membranes, Transport, and Cellular Homeostasis

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Chapter overview

Every living cell must maintain an internal environment that is different from its surroundings. A cell needs useful nutrients, must remove wastes, and must keep water, ions, and other molecules within workable ranges. The plasma membrane makes this control possible. It forms a boundary between the cytoplasm and the extracellular environment, but it is not a sealed wall. Instead, it is a selective, responsive structure that controls movement, receives signals, and connects the cell to its neighbors.

This chapter explains how membrane structure supports membrane function. It then compares passive transport, active transport, and vesicular transport. The final sections connect these processes to homeostasis in animal and plant cells.

1. The plasma membrane is a selective boundary

The plasma membrane surrounds the cell and separates the cytoplasm from the extracellular fluid. This separation allows the cell to regulate chemical reactions and maintain conditions that may be very different from those outside the cell. For example, many animal cells maintain a high concentration of potassium ions inside the cell and a high concentration of sodium ions outside the cell.

The membrane is described as **selectively permeable** because some substances cross it more easily than others. Small nonpolar molecules such as oxygen and carbon dioxide can pass through the lipid portion of the membrane relatively easily. Ions and most large or polar molecules cannot move freely through the membrane's hydrophobic interior. They usually require transport proteins.

Selective permeability supports **homeostasis**, the maintenance of relatively stable internal conditions despite changes inside or outside an organism. Stability does not mean that conditions never change. A cell constantly exchanges matter and energy with its surroundings. Homeostasis is dynamic: transport processes adjust continuously to keep variables within a useful range.

2. The phospholipid bilayer

Most of the membrane is built from phospholipids. A phospholipid has a polar, hydrophilic head and two nonpolar, hydrophobic fatty-acid tails. "Hydrophilic" means attracted to water, while "hydrophobic" means avoiding interactions with water.

In an aqueous environment, phospholipids arrange themselves into a bilayer. The hydrophilic heads face the watery cytoplasm and extracellular fluid. The hydrophobic tails point inward toward one another, away from water. This arrangement is stable because it allows the polar heads to interact with water while shielding the nonpolar tails.

The bilayer's interior creates a barrier to ions and most polar molecules. These substances interact well with water but not with the membrane's nonpolar core. In contrast, small nonpolar molecules dissolve in the lipid region and cross more readily. Thus, the chemical properties of phospholipids create the basic pattern of membrane permeability.

Phospholipids are not locked in place. Many move sideways within their own layer, which makes the membrane flexible. The membrane can bend, fuse with vesicles, and change shape without breaking apart. This flexibility is important during cell movement, cell division, endocytosis, and exocytosis.

3. The fluid mosaic model

Scientists use the **fluid mosaic model** to describe membrane organization. "Fluid" refers to the movement of phospholipids and many proteins within the membrane. "Mosaic" refers to the mixture of proteins, carbohydrates, lipids, and other components arranged throughout the bilayer.

Temperature affects membrane fluidity. At low temperatures, phospholipid movement slows and membranes may become more rigid. At high temperatures, movement increases and the membrane may become too fluid. The fatty-acid tails also matter. Unsaturated tails contain one or more double bonds that create bends, preventing neighboring phospholipids from packing tightly. Membranes with more unsaturated tails generally remain more fluid at lower temperatures than membranes with many saturated tails.

Cholesterol helps stabilize animal-cell membranes. At moderate and high temperatures, cholesterol restrains phospholipid movement and reduces excessive fluidity. At low temperatures, it prevents phospholipids from packing too closely, helping the membrane remain fluid. Cholesterol therefore acts as a fluidity buffer rather than simply making the membrane more or less fluid in all conditions.

4. Membrane proteins and carbohydrates

Membrane proteins perform much of the membrane's specialized work. **Integral membrane proteins** are embedded in the bilayer. Many extend all the way across it and are called transmembrane proteins. **Peripheral proteins** attach loosely to the membrane surface or to other proteins.

Transport proteins help substances cross the membrane. **Channel proteins** create hydrophilic passageways through which specific ions or molecules can move. Some channels are always open, while gated channels open or close in response to a signal. **Carrier proteins** bind specific substances and change shape to move them across the membrane.

Other membrane proteins act as receptors. A signaling molecule binds to a receptor because its shape and chemical properties complement a specific binding site. Binding changes the receptor and starts a response inside the cell. The signaling molecule does not always enter the cell; the receptor can transmit information across the membrane.

Some proteins function as enzymes, anchors, or identity markers. Anchoring proteins attach the membrane to the cytoskeleton inside the cell or to the extracellular matrix outside it. These connections help maintain cell shape and organize proteins. Glycoproteins, proteins with attached carbohydrate chains, help cells recognize one another. Glycolipids also carry carbohydrate chains that contribute to cell identity and communication.

5. Concentration gradients and diffusion

A **concentration gradient** exists when the concentration of a substance differs between two regions. Molecules in liquids and gases are in constant random motion. Although any individual molecule may move in any direction, a population of molecules shows a net movement from an area of higher concentration to an area of lower concentration. This net movement is **diffusion**.

Diffusion does not require the cell to supply metabolic energy. It is driven by the random motion of particles and the existence of a gradient. For this reason, diffusion is a type of passive transport. A substance diffuses down its concentration gradient.

At **dynamic equilibrium**, particles continue to move, but there is no net change in concentration. Equal numbers of particles cross in each direction over time. Equilibrium therefore does not mean that molecular motion stops.

Several factors affect diffusion rate. A steeper concentration gradient usually increases net diffusion. Higher temperature increases particle motion. A larger membrane surface area provides more space for movement. A shorter diffusion distance also increases the rate. The size and chemical properties of a substance affect how easily it crosses a particular membrane.

6. Simple diffusion through the bilayer

In **simple diffusion**, a substance moves directly through the phospholipid bilayer without a transport protein. Small nonpolar molecules are the best candidates. Oxygen enters many cells by simple diffusion because cells use oxygen during cellular respiration, often keeping its concentration lower inside than outside. Carbon dioxide produced by respiration often diffuses out of cells down its concentration gradient.

Simple diffusion is selective even though no protein chooses the molecules. Selection results from the structure of the bilayer. Charged ions are surrounded by water molecules and face a strong energetic barrier in the hydrophobic interior. Large polar molecules also cross poorly. Their inability to diffuse rapidly does not mean they cannot cross at all; cells often provide protein pathways for them.

The direction of simple diffusion depends on the gradient. A membrane does not actively pull oxygen into every cell. If oxygen concentration became higher inside than outside, the net direction would reverse. Passive transport always depends on the existing gradient.

7. Facilitated diffusion

Facilitated diffusion moves a substance down its concentration or electrochemical gradient through a membrane protein. Like simple diffusion, it does not directly require cellular energy. Unlike simple diffusion, it depends on specific channels or carriers.

Ion channels allow selected ions to cross. Specificity may depend on the channel's diameter, charge, and binding sites. A potassium channel favors potassium ions over sodium ions even though both are positively charged. Gated channels provide additional control. A voltage-gated channel responds to a change in membrane voltage, a ligand-gated channel responds to chemical binding, and a mechanically gated channel responds to stretching or pressure.

Carrier proteins also support facilitated diffusion. A carrier binds a molecule on one side, changes shape, and releases it on the other side. Glucose transporters in many animal cells operate this way. Because a cell has a limited number of carriers, the transport rate can reach a maximum when all carriers are occupied. This saturation differs from simple diffusion through the lipid bilayer.

Facilitated diffusion cannot move a substance against its gradient. A protein pathway may make movement possible and regulate its rate, but the gradient still supplies the driving force.

8. Osmosis: the diffusion of water

Osmosis is the net movement of water across a selectively permeable membrane. Water moves from a region with more free water and lower solute concentration toward a region with less free water and higher solute concentration, assuming

the solute cannot cross. The net movement continues until the difference is balanced or pressure prevents further movement.

Water can cross the lipid bilayer slowly. Many cells also contain **aquaporins**, channel proteins that greatly increase water movement. Aquaporins are selective for water and allow cells to adjust rapidly to osmotic changes.

It is useful to focus on water movement rather than saying that a solute "pulls" water with intention. Molecules have no goals. Random water movement occurs in both directions, but a difference in water availability produces net movement.

Osmosis can change cell volume. If water enters faster than it leaves, a cell swells. If water leaves faster than it enters, the cell shrinks. The outcome depends on the relative concentrations of nonpenetrating solutes on the two sides of the membrane.

9. Tonicity and animal cells

Tonicity describes how a surrounding solution affects a cell's volume. It depends mainly on nonpenetrating solutes, because solutes that cross the membrane can redistribute and may not produce a lasting osmotic effect.

In an **isotonic solution**, the concentration of nonpenetrating solutes is effectively equal inside and outside the cell. Water moves in both directions at equal rates, so there is no net change in cell volume.

In a **hypotonic solution**, the surrounding fluid has a lower concentration of nonpenetrating solutes than the cell. Water enters the cell by osmosis. An animal cell may swell and can burst, a process called lysis, if water entry is extreme.

In a **hypertonic solution**, the surrounding fluid has a higher concentration of nonpenetrating solutes than the cell. Water leaves the cell, causing it to shrink. In red blood cells, this shrinkage is called crenation.

The terms isotonic, hypotonic, and hypertonic describe one solution relative to a particular cell or solution. A liquid is not universally "hypertonic" by itself. Its effect depends on the membrane and the concentrations of relevant nonpenetrating solutes.

10. Osmosis in plant cells

Plant cells respond differently to osmotic conditions because a rigid cell wall surrounds the plasma membrane. In a hypotonic environment, water enters the cell. The expanding plasma membrane presses against the wall, and the wall resists further expansion. The resulting **turgor pressure** helps support nonwoody plant tissues. A plant cell with strong turgor pressure is described as turgid.

In an isotonic environment, a plant cell loses some turgor pressure and becomes flaccid. The plant may wilt because its cells provide less structural support. Isotonic conditions are therefore not necessarily ideal for plant tissue even though they are often appropriate for animal cells.

In a hypertonic environment, water leaves the plant cell. The plasma membrane may pull away from the cell wall, a process called **plasmolysis**. Severe plasmolysis disrupts normal cell function.

The cell wall does not stop osmosis. It limits expansion and produces pressure that opposes continued water entry. Water still crosses the plasma membrane according to water potential and osmotic conditions.

11. Electrochemical gradients

Ions are influenced by both concentration and electrical charge. The combined effect is called an **electrochemical gradient**. For example, a positively charged ion may be chemically driven into a cell because its concentration is lower inside, but an electrically positive interior would oppose its entry. The net direction depends on both forces.

Separation of charge across a membrane produces a **membrane potential**, a voltage difference between the two sides. Many cells have a negative interior relative to the extracellular fluid. Membrane potential affects ion movement and is especially important in neurons and muscle cells.

When an ion moves through an open channel, it travels down its electrochemical gradient. No direct energy input is needed for that movement. However, cells often spend energy to create and maintain the gradients that later drive passive ion flow.

12. Active transport

Active transport moves substances against their concentration or electrochemical gradients. Because this movement is energetically unfavorable, it requires an energy source and a transport protein.

In **primary active transport**, a pump directly uses energy, commonly from ATP hydrolysis. The protein binds its transported substances, receives energy from ATP, changes shape, and releases the substances on the opposite side. Pumps are specific, and their repeated action can build steep gradients.

Active transport is necessary because diffusion alone would eventually eliminate concentration differences. Cells instead maintain nonequilibrium conditions. These stored gradients can support electrical signals, nutrient uptake, pH regulation, and cell volume control.

A pump does not make every molecule move in one direction at once. It cycles through binding and release steps. Its rate depends on available ATP, the number of pumps, and the concentrations of the transported substances.

13. The sodium-potassium pump

The sodium-potassium pump is a primary active transporter in animal cells. In each cycle, it moves three sodium ions out of the cell and two potassium ions into the cell while using energy from one ATP. Both ions move against their concentration gradients.

The cycle begins when three sodium ions bind inside the cell. ATP transfers a phosphate group to the pump, causing its shape to change. The sodium ions are released outside. Two potassium ions then bind from the extracellular side. Release of the phosphate returns the pump toward its original shape, and potassium is released inside.

Because three positive charges leave for every two that enter, the pump makes a small direct contribution to membrane potential. More broadly, it maintains sodium and potassium gradients that are essential for nerve impulses, muscle contraction, and secondary active transport.

If ATP production stops, the pump cannot maintain these gradients indefinitely. Ion distributions slowly shift, water balance can be disrupted, and cell function declines.

14. Secondary active transport

Secondary active transport uses energy stored in an ion gradient rather than directly using ATP at the cotransporter. A primary pump first creates the gradient. When the ion later moves down that gradient, its favorable movement can power another substance moving against its own gradient.

In **symport**, the driving ion and the transported substance move in the same direction. The sodium-glucose cotransporter in intestinal and kidney cells uses sodium movement into the cell to bring glucose into the cell against its glucose concentration gradient.

In **antiport**, the substances move in opposite directions. For example, the downhill movement of one ion into a cell may drive another ion out. Symporters and antiporters are both cotransport proteins.

Secondary active transport is still considered active because at least one substance moves against its gradient. Its energy connection is indirect: ATP powers the primary pump, the primary pump builds an ion gradient, and the cotransporter uses that gradient.

15. Bulk transport with vesicles

Large particles, macromolecules, and large quantities of material may be transported in membrane-bound vesicles. Vesicular transport requires cellular energy and depends on the membrane's ability to bend and fuse.

In **endocytosis**, the plasma membrane folds inward around material and pinches off to form an internal vesicle. In **phagocytosis**, sometimes called cell eating, a cell engulfs a large particle such as a bacterium. In **pinocytosis**, or cell drinking, a cell takes in extracellular fluid and dissolved substances.

Receptor-mediated endocytosis is more selective. Specific molecules bind to receptors concentrated in a region of the membrane. The membrane then forms a vesicle containing the bound molecules. This process allows cells to collect substances that may be present at low concentrations.

In **exocytosis**, an internal vesicle fuses with the plasma membrane and releases its contents outside the cell. Neurons release neurotransmitters by exocytosis, and gland cells release protein hormones and digestive enzymes in the same general way. Exocytosis also adds the vesicle membrane to the plasma membrane, while endocytosis removes membrane from the surface.

16. Comparing the major transport mechanisms

Simple diffusion, facilitated diffusion, and osmosis are passive. They move substances down an existing gradient and do not directly consume cellular energy. Simple diffusion uses the lipid bilayer, while facilitated diffusion uses channels or carriers. Osmosis specifically describes water movement.

Primary active transport uses an energy source such as ATP to move substances against a gradient. Secondary active transport uses the energy stored in another substance's gradient. Endocytosis and exocytosis move material in vesicles and require energy.

Transport proteins do not automatically make a process active. A channel used for facilitated diffusion is a protein, but the transported substance still moves downhill. The defining question is whether the transported substance moves down its gradient or whether energy is used to move it uphill.

The same substance can use different mechanisms in different cells or conditions. Water may cross slowly through the bilayer or rapidly through aquaporins. Glucose may enter one cell by facilitated diffusion and enter another by sodium-glucose cotransport.

17. Transport and cellular homeostasis

Membrane transport processes work together rather than in isolation. Consider an intestinal epithelial cell. The sodium-potassium pump uses ATP to keep intracellular sodium low. Sodium then moves back into the cell through a

sodium-glucose symporter, bringing glucose with it from the intestinal space. Glucose exits the other side of the cell through a facilitated-diffusion carrier and enters the blood. Water may follow solute movement by osmosis.

This example shows an energy chain. ATP does not directly power the glucose symporter, but ATP maintains the sodium gradient that makes cotransport possible. If the sodium-potassium pump stops, the sodium gradient weakens and glucose uptake by the symporter declines.

Cells regulate transport by changing the number, location, or activity of membrane proteins. Signals may open channels, insert transporters into the membrane, or remove them by endocytosis. Regulation lets a cell respond to changing conditions without rebuilding the entire membrane.

Transport failures can disturb homeostasis. A defective channel may alter ion balance. Loss of pump activity may collapse gradients. Extreme osmotic conditions may change cell volume. Because transport affects water, nutrients, wastes, electrical signals, and pH, membrane function is central to cell survival.

18. Experimental reasoning about membranes

Biologists often investigate transport by changing one variable and observing movement. Suppose cells are placed in solutions with different concentrations of a nonpenetrating solute. A decrease in cell mass indicates net water loss, while an increase indicates net water gain. The solution that produces no net mass change is approximately isotonic to the cells.

To test whether a substance uses a protein transporter, researchers can compare transport rates at increasing concentrations. A carrier-mediated process often approaches a maximum rate because the number of carriers is limited. Simple diffusion is less likely to show the same saturation pattern under comparable conditions.

Energy dependence can be tested by reducing ATP production. If transport rapidly declines when ATP becomes unavailable, the mechanism may depend directly or indirectly on active transport. Interpretation must be careful because loss of ATP affects many cell functions at once.

Specific inhibitors provide additional evidence. If blocking the sodium-potassium pump reduces sodium-glucose cotransport, the result supports the idea that the sodium gradient powers glucose uptake. The inhibitor did not have to bind the cotransporter itself; disrupting the upstream gradient is enough to affect the downstream process.

Chapter summary

The plasma membrane is a fluid phospholipid bilayer containing proteins, carbohydrates, and cholesterol. Its hydrophobic interior makes it selectively permeable. Small nonpolar molecules can cross by simple diffusion, while many ions and polar molecules require specific proteins.

Passive transport moves substances down concentration or electrochemical gradients. It includes simple diffusion, facilitated diffusion, and osmosis. Tonicity predicts how nonpenetrating solutes affect cell volume. Animal cells may lyse or shrink, while plant cell walls create turgor pressure and limit expansion.

Active transport maintains gradients. Primary pumps use energy such as ATP directly. Secondary transporters use energy stored in gradients created by primary pumps. The sodium-potassium pump moves three sodium ions out and two potassium ions in per ATP, supporting membrane potential and many other transport processes.

Endocytosis brings material into cells in vesicles, and exocytosis releases material. Together, all of these mechanisms let cells acquire nutrients, remove waste, communicate, regulate water and ions, and maintain homeostasis.

Key terms

- active transport
- carrier protein
- concentration gradient
- electrochemical gradient
- exocytosis
- fluid mosaic model
- hypertonic
- isotonic
- osmosis
- phospholipid bilayer
- secondary active transport
- sodium-potassium pump
- turgor pressure
- aquaporin
- channel protein
- diffusion
- endocytosis
- facilitated diffusion
- homeostasis
- hypotonic
- membrane potential
- passive transport
- primary active transport
- selective permeability
- tonicity